HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
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Final Report
Escrow Payment System (IPS)

Course: System Analysis and Design – IT3120E
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Group 1

Chapter 1

Introduction

1.1 Problem Definition

Today, online transactions are part of almost every digital service – from e-commerce and freelancing platforms to cross-border trade and marketplace payments. However, many transactions still face a fundamental problem: the lack of trust between buyers and sellers who do not know each other. In a typical online transaction, the buyer must pay before receiving the goods or services, while the seller must deliver before receiving payment. This creates a high risk of fraud, non-delivery, late delivery, or payment reversal.

In addition, traditional payment systems are designed to move money quickly, but they do not support conditional release. Once a payment is authorized, funds are transferred immediately to the seller, leaving the buyer unprotected if the goods are not delivered as agreed. Similarly, sellers may be exposed to non-payment risk or chargeback fraud. When disputes arise, resolving them often requires manual intervention, chargebacks, or legal action, which is slow, expensive, and damaging to trust.

These challenges are even more serious for high-value and cross-border transactions, where currency conversion, different regulations, and extra transaction steps increase complexity and cost. In markets like Vietnam and Southeast Asia, existing escrow solutions are often limited to large institutions, law firms, or specific marketplaces, making them inaccessible to small and medium-sized enterprises (SMEs), freelancers, and individual buyers.

To solve these problems, a more efficient and trustworthy payment structure is needed – one that can hold funds securely, release them only when predefined conditions are met, and resolve disputes fairly. The proposed **Escrow Payment System (EPS)** is designed to meet this need. It acts as a trusted third party between the payer and the receiver, helping to verify, hold, and release payments in a fast, secure, and transparent way.

1.2 Proposed Solution

The Escrow Payment System offers a modern and flexible solution for managing conditional payments. Instead of transferring funds directly from the buyer to the seller, it creates a single platform that holds the funds in a secure escrow account and automates the escrow lifecycle. The goal is to make high-value and cross-border transactions safer, more reliable, and easier to track.

Key features of the proposed system include:

- **Conditional Release:** A rules engine that releases funds only when predefined conditions are met, such as buyer confirmation, logistics delivery verification, or milestone approval.
- **Dispute Resolution:** A structured workflow for evidence submission, mediation, and arbitration to resolve conflicts fairly without litigation.
- **Centralized Escrow Management:** A single dashboard to create, fund, and monitor transactions, with the ability to handle complex flows involving multiple parties (buyers, sellers, and brokers).
- **Interoperability & Verification:** Seamless integration with banks, e-wallets, KYC/AML providers, and third-party logistics via APIs to automatically confirm identity and delivery.
- **Cross-Border Support:** Multi-currency handling, automated currency conversion, and compliance with regional financial regulations.
- **Enhanced Security:** Strong encryption, tokenization, multi-factor authentication, and end-to-end traceability for every transaction.
- **Analytics Dashboard:** Tools to view escrow volume, dispute rates, and release times to help administrators monitor system health.

By holding funds securely, releasing them conditionally, and providing a fair dispute resolution mechanism, the Escrow Payment System helps organizations and users reduce fraud, build trust, and gain confidence in their online financial operations. their online financial operations.

Chapter 2

Planning

2.1 Motivation

With the rise of online transactions, users expect fast, secure, and reliable payments. However, existing systems still suffer from delays, high fees, and poor coordination between buyers, sellers, and platforms. More importantly, they do not address the trust gap: buyers risk paying for goods that are never delivered, while sellers risk delivering goods that are never paid for. These problems slow down transactions, increase fraud, and make high-value or cross-border deals especially difficult.

The proposed Escrow Payment System aims to solve these issues by acting as a trusted bridge between payers and receivers. It holds funds securely in an escrow account, releases them only when predefined conditions are met, and provides a fair dispute resolution mechanism. The system automates verification, ensures regulatory compliance, and provides faster, safer, and more transparent transactions, especially in high-value and cross-border scenarios

2.2 Initial System Request

2.2.1 Business Need

Businesses and individuals turn to escrow payment systems to fulfill a range of essential operational requirements:

- **Building Trust Between Strangers:** Protect buyers and sellers who do not know each other by holding funds in a neutral account until conditions are met.
- **Supporting High-Value Transactions:** Provide a safeguard for large purchases such as real estate, vehicles, or bulk orders where both parties need extra protection.

- **Enabling Cross-Border Trade:** Provide a neutral ground for international transactions involving different currencies, regulations, and legal systems.
- **Reducing Disputes and Chargebacks:** Automate verification and provide a structured dispute workflow to reduce the cost of chargebacks and legal action.
- **Supporting Marketplaces and Platforms:** Allow e-commerce and freelancing platforms to embed escrow functionality and reduce platform liability.
- **Automating Financial Workflows:** Automate the escrow lifecycle from funding to conditional release, dispute handling, and reporting.

2.2.2 Business Requirements

When choosing an escrow provider, businesses must evaluate a number of key requirements:

- **Security and Compliance:** Compliance with escrow licensing, PCI DSS, KYC/AML, and GDPR regulations.
- **Segregated Accounts and Auditability:** Escrow funds held in segregated accounts with immutable audit logs for every state change.
- **Conditional Release Engine:** Flexible release conditions such as buyer confirmation, delivery verification, milestone approval, or time-based release.
- **Dispute Resolution Workflow:** A structured process for evidence submission, mediation, and arbitration.
- **Integration Capabilities:** Seamless integration with e-commerce platforms, accounting software, logistics providers, and banks via APIs.
- **Scalability and Reliability:** Handle growing escrow volume without compromising performance or uptime.
- **Reporting and Analytics:** Detailed reporting on escrow volume, release times, dispute rates, and compliance metrics.
- **Cost and Fee Structure:** Transparent and competitive pricing covering escrow, transaction, and currency conversion fees.
- **Customer Support:** Responsive multi-channel support for dispute and payment issues.
- **International Payment Capabilities:** Multi-currency support with clear foreign exchange rates and regional compliance.

- **Dispute Resolution Workflow:** A structured process and communication portal for evidence submission, mediation, and arbitration.

2.2.3 Business Value

The proposed Escrow Payment System will deliver value through:

- **Improved Trust:** A neutral escrow account protects both parties and reduces fraud.
- **Streamlined Processes:** Automated verification and conditional release replace manual paperwork and reduce operational delays.
- **Enhanced Security:** Segregated accounts, encryption, and audit logs protect funds and data.
- **Cost Savings:** Fewer disputes, chargebacks, and legal costs.
- **Better User Experience:** A single platform that simplifies the escrow lifecycle.
- **Market Expansion:** Cross-border and multi-currency support opens new markets.

2.2.4 Special Issues or Constraints

- **Regulatory Compliance:** Must obtain escrow licenses and comply with central bank regulations in all operating regions.
- **Segregated Account Management:** Requires careful banking partnerships and strict internal controls.
- **Scalability:** Must handle high concurrent escrow volume during peak periods.
- **Security:** Strong encryption, MFA, and fraud detection are essential.
- **Integration:** Smooth interoperability with banks, gateways, logistics, and platforms.
- **Dispute Resolution Fairness:** Transparent rules for evidence submission and arbitration, requiring a dedicated operational team or AI-assisted moderation to handle edge cases.
- **Usability:** Simple interface for buyers, sellers, and administrators.
- **Cross-Border Complexity:** Multiple currencies, regulations, and legal systems add significant complexity.

2.3 Feasibility Analysis

2.3.1 Technical Feasibility

Technical Resources: Modern web and cloud technologies provide the necessary infrastructure for secure, high-performance escrow systems. EPS can be implemented using:

- **Frontend:** ReactJS or Angular for responsive web interfaces.
- **Backend:** Node.js, Django, or Spring Boot for API and escrow logic.
- **Database:** PostgreSQL or MongoDB for storing escrow records and audit logs.
- **Cloud Infrastructure:** AWS, Azure, or Google Cloud for scalability, load balancing, and disaster recovery.
- **Security Frameworks:** AES-256 encryption, SSL/TLS, OAuth 2.0, and multi-factor authentication.
- **Escrow-Specific:** Segregated account integration with banking partners, an automated rules engine for conditional release, and logistics/KYC API connectors.

Technical Expertise: Development skills – API integration, database management, encryption, financial data processing, and escrow compliance.

Risk Assessment: Three main types of risk have been identified: time, resource, and functional, with additional escrow-specific risks around regulatory compliance and banking partnerships.

2.3.2 Operational Feasibility

- **User Readiness:** Banks, fintech firms, marketplaces, and users are already familiar with escrow concepts and online payment platforms.
- **Organizational Impact:** Enhances workflow efficiency by automating escrow creation, conditional release, and dispute routing. The system separates technical operations from business operations, ensuring clear ownership of each function.
- **Maintenance and Support:** Ongoing operations are handled by two distinct teams:
  - A **Technical Team** responsible for system uptime, security monitoring, bug fixes, and infrastructure maintenance.
  - An **Operations & Compliance Team** responsible for dispute resolution, KYC/AML reviews, customer support, and regulatory reporting.

2.3.3 Economic Feasibility

Cost of Development:

- **Initial Setup:** $\$120{,}000 - \$150{,}000$, covering backend and frontend development, escrow account integration, API integration, UI design, security implementation, testing, and deployment.
- **Recurring Costs:** $\$4{,}000 - \$7{,}000$ per month, including cloud hosting, segregated account maintenance, compliance monitoring, and support.

Revenue Generation:

- **Escrow Fees:** A commission $(0.5\% - 1.5\%)$ per successful escrow transaction. This is higher than standard payment gateways because escrow provides greater protection, longer fund-holding periods, and dispute handling.
- **Subscription Model:** Monthly or annual plans for marketplaces, SMEs, and financial institutions.
- **Value-Added Services:** Premium features such as dispute arbitration, real-time analytics, and custom API integrations.

Cost Efficiency: Automating escrow lifecycle, verification, and dispute routing significantly reduces manual intervention and legal costs.

Return on Investment (ROI): Break-even point within 2–3 years, supported by escrow fees, subscription revenue, and expanding transaction volume.

2.3.4 Legal & Regulatory Feasibility

Compliance:

- **Laws and Regulations:** In Vietnam, there is no standalone "escrow license." The system must obtain relevant **intermediary payment licenses** from the State Bank of Vietnam (SBV) and strictly maintain **Payment Guarantee Accounts** (Tài khoản đảm bảo thanh toán) with partnered commercial banks to hold user funds legally.
- **Third-party Service:** Contracts with banking partners, logistics providers, and KYC/AML vendors must be carefully managed.

Data Privacy: Robust security measures and compliance with PCI DSS, GDPR, and Vietnamese regulations on Non-Cash Payments and SBV guidance. Includes encryption, strong authentication, segregated accounts, and transparent terms of service.

2.3.5 Market Feasibility

Demand: Rapid growth of digital commerce, cross-border trade, and high-value online transactions has created strong demand for secure escrow solutions.

Competitive Landscape: Many existing escrow providers are tied to specific banks, law firms, or marketplaces, with high fees and limited interoperability. An independent, API-first, and cost-effective escrow platform can fill a significant market gap and establish a strong competitive advantage.

Target Market: SMEs, online marketplaces, freelancers, cross-border traders, and financial institutions seeking flexible escrow integration.

Market Potential: Adoption rates for escrow-backed online payments are expected to rise significantly, driven by government initiatives promoting digital transformation and financial inclusion.

2.4 Initial Project Plan

2.4.1 Project Scope Estimation

Main Objective: Develop a Minimum Viable Product (MVP) for the Escrow Payment System (EPS) that supports basic escrow flows (create → fund → hold → release/refund), integration with multiple payment methods, conditional release rules, and basic dispute resolution. The MVP will also include an administrative dashboard and APIs for merchants and marketplaces.

Functional Scope (MVP):

- User authentication and registration (buyer, seller, admin).
- Escrow creation, funding, and tracking.
- Conditional release engine with configurable rules.
- API connections to at least one bank and one payment gateway (mock or real).
- Basic dispute workflow with evidence submission.
- Logging, audit trail, and basic analytics dashboard.
- Security mechanisms: OAuth2, TLS encryption, MFA, and segregated account integration.

Estimated Timeline (21 weeks total):

- Requirement Analysis: 2 weeks

- System & UI/UX Design: 3 weeks
- Backend, Escrow Logic & API Development: 7 weeks
- Frontend and Dashboard Development: 5 weeks (in parallel with backend)
- Dispute Workflow & Integration: 2 weeks
- Testing, Security Review, Penetration Testing, and Deployment: 2 weeks

The Testing phase is extended to 2 weeks because escrow systems hold user funds for extended periods and enforce complex disbursement rules. Security review and penetration testing are critical to prevent financial loss and regulatory violations. The total timeline is 21 weeks, with any remaining buffer absorbed into the extended testing phase.

Deliverables:

- Requirement Specification Document
- System Architecture and Database Design
- API Specification
- Frontend & Backend Source Code
- Escrow Condition & Dispute Policy Document
- Test Report and Deployment Guide

2.4.2 Staffing Plan

For a small company with fewer members, the team structure must remain lean yet cover all critical roles required for developing, securing, and maintaining the Escrow Payment System (EPS):

- **Project Manager / Business Analyst:** Oversees project planning, requirement gathering, and stakeholder coordination. Ensures deliverables align with business goals and manages the overall project timeline. Works closely with external Legal/Compliance Consultants to define dispute resolution frameworks and ensure central bank compliance.
- **Technical Lead:** Defines system architecture, technology stack, and coding standards. Oversees integration, code reviews, and ensures system scalability, performance, and security compliance.

- **Full-Stack Developer (Backend + Frontend):** Implements backend APIs and frontend dashboards. Handles database operations, escrow logic, and UI functionalities.
- **DevOps & Security Engineer:** Manages CI/CD pipelines, deployment, and infrastructure monitoring. Also responsible for implementing security policies, SSL/TLS setup, and disaster recovery.
- **QA Engineer / UI-UX Designer:** Performs testing (functional, integration, load) and ensures KYC/AML, escrow compliance, and data protection standards are met. Assists with UI/UX improvements.
- **Operations & Compliance Specialist (part-time or shared):** Handles dispute resolution, KYC/AML reviews, customer support, and regulatory reporting. Ensures the platform operates within SBV guidelines and maintains Payment Guarantee Accounts properly.

2.4.3 CASE Tools

Modeling and Design: StarUML / Enterprise Architect for use case, class, and sequence diagrams. Figma for UI wireframes and collaborative design.

Project Management: Jira for backlog and sprint tracking.

Development and CI/CD: Git + GitHub/GitLab for version control. GitHub Actions / GitLab CI for build, test, and deploy automation. Docker for containerization; Kubernetes for orchestration if needed.

Database and API Tools: PostgreSQL. Postman / Swagger (OpenAPI) for API testing and documentation.

Testing and Security: Pytest / Jest for unit and integration testing. SonarQube / Dependabot for code quality and dependency security scanning. k6 / JMeter for performance and load testing.

Monitoring and Logging: ELK Stack (Elasticsearch, Logstash, Kibana) for centralized logging. Prometheus + Grafana for system monitoring and alerts.